

Bio331 Fall 2025 Syllabus

Bio331: Computational Systems Biology

In the era of high-throughput genomics, proteomics, and transcriptomics, we can now computationally explore biological processes at a systems level. Networks, or graphs, have become a dominant mathematical representation in this area of research. We will draw on the vast amount of established graph theory to learn about network models currently applied to biological systems.

About this syllabus

This syllabus is a public, searchable document that contains all of the details for this iteration of the course. It serves as a contract for the students taking Bio331 - if you take this course, you agree to abide by the policies described here. As the instructor, I commit to following this syllabus and administering it fairly and equitably. The main parts of the syllabus are:

- Goals: the course goals and learning objectives.
- Types of Work: an explanation of how components in this course are evaluated and contribute to the final course grade.
- Support: the support systems offered in the course.
- Policies: policies surrounding attendance, deadlines, collaboration, and online resources.
- Learning Environment: my commitment to making the classroom an inclusive learning environment.
- Keys for Success: tips for succeeding in the course.

The final version of this syllabus will be set by the first day of class. Any modifications during the semester will be noted in the change log and will be communicated to the class via Moodle.

Course details

Instructor: Anna Ritz (Biology 200B, see the support page for student hours)

Lecture: MWF 10-10:50am in ETC 211

Lab: M 1:10-4:00 in ETC 208

Materials: All reading, videos, and other materials will be freely available via Moodle. A week-by-week schedule is available on Moodle as well.

Communication & Technology

Official communications about all course materials and assignments will be done through the Moodle page.

- Check your email every day to stay updated.
- We will use GitHub for the programming assignments.
- We will use Moodle for written assignment submissions.
- We will use other technologies (like Padlet or ImageJ/FIJI) for class activities.
- If you have a question, others likely have the same question. Reach out at any time to me at aritz@reed.edu.

The Honor Principle is in effect at all times. This includes our work and interactions in class as well as our work done outside of class. We will have a discussion at the beginning of the semester about how the honor principle might be relevant in Bio131.

Goals & Objectives

At the end of Bio331, you will be able to:

1. Represent biological systems as networks and understand how they are built from data.
2. Implement graph algorithms and apply them to biological networks (including simulated and real-world datasets).
3. Interpret the results of an algorithm run on a biological network.
4. Use technologies that enable collaboration and communication.
5. Learn how to work through a challenging research project and mark progress along the way.
6. Communicate work to a broad audience.

Bio331 satisfies the Group III and Group III-Data Collection and Analysis distribution requirements, and you can find the relevant group distribution learning outcomes in the course catalog description.

Course support

Bio331 has a number of support systems to help you in this class.

Student Hours: Wednesdays 11-12pm and 3-4pm

Student hours are times where I am in my office (Biology 200B) waiting for students to come talk with me. Drop in during these times to talk about anything

(class related or otherwise), no appointment needed. Email me to set up an appointment if you cannot make student hour times.

Tutoring (DoJo)

Tutoring is offered by Academic Support Services, and gives students additional times outside of class to get help with assignments and the course material. Each student can get up to one hour per week *per class* of individual (one-on-one) tutoring. If someone requests individual tutoring, I will work to find a student who can tutor Bio331 material. Please reach out to the Office of Academic Support and Cc me if you are interested in tutoring.

Academic Accommodations

It is my goal to create a learning experience that is as accessible as possible. If you anticipate any issues related to the format, materials, or requirements of this course, please meet with me outside of class so we can explore potential options.

Please discuss any documented accommodations with me. Disability Accommodation Notification Letters can be obtained from Disability & Accessibility Resources. All discussions will remain confidential.

Illness and Exceptional Circumstances

There may be cases where you cannot attend class for stretches at a time (due to illness, for example). In this case, email me to strategize a successful plan for learning the material. If you miss multiple classes and do not email me, I may reach out to make sure you are OK.

In rare cases, there may be circumstances beyond your control where you are unable to meet the expectations outlined in this syllabus. If there are extenuating circumstances that might impact your academic performance in this class, please inform me as soon as you can to strategize a plan for completing all parts of the course. Additionally, you may want to reach out to the Student Life Office at student-life@reed.edu, phone: (503) 517-7396, office: Eliot 218. The Student Life Office is dedicated to assisting students in accessing campus and support resources, especially during challenging situations.

Obligated Reporting

I am happy to talk with you about anything that concerns you, but note that I (and many faculty) are obligated reporters who must report possible violations of the Title IX and DHSM policies which govern discrimination and harassment on the basis of sex, gender, and gender identity. Confidential resources are available at Health & Counseling Services and through the SHARE program; more information is available on this page.

Course Policies

There are four policies for this course: one for attendance, one for deadlines, one for collaboration (which is encouraged for the programming assignments) and one for the use of online resources and generative AI (which will be discussed in class).

Attendance Policy

Why Attend? Attending the lectures and completing work during labs will put you in the best position to succeed in the course. Lectures will provide valuable information about the assignments, and labs will include some group work. Labs also offer time for me to provide guidance - they could be framed as small-group office hours working through problems together.

What's the Policy? Email me if you cannot attend lecture or lab. You are expected to review the missed material and meet with me if anything is unclear.

There are three classes with activities and 14 labs where your engagement is expected. If you have spoken with me about not being able to attend, you will be able to catch up with this work asynchronously.

You are expected to attend the presentations during Finals Week.

There are some religious holidays and other events in the fall that might affect student attendance - please note these in your calendars and let me know if your attendance will be affected.

- September 22-24: Rosh Hashanah
- October 1-2: Yom Kippur
- October 20: Diwali begins
- November 11: Veterans Day
- November 27: Thanksgiving Day

If you miss multiple classes, I may reach out to make sure you are OK. Communication is key. Please see “Illness and Exceptional Circumstances” on the support page for more information about extended absences.

The Main Takeaway: Attending and participating is expected and will set you up for success.

Deadline Policy

The programming assignments, quizzes, and the research project assignments all have deadlines. The purpose of these deadlines is for you to complete the assignments at a steady pace throughout the semester.

Why Submit Work on Time? The programming assignments and take-home quizzes are designed so that you are working on them after you have learned the

relevant material in lecture and lab. If you are working on older assignments when another assignment is out, you might lose ground with the current coursework.

What's the Policy? You must submit whatever work you have by the deadline. Any work that is marked as “Missing” or “Partially Complete” can be resubmitted after receiving feedback from me.

- The deadline to resubmit programming assignments is Mon 11/10 (3 weeks after the last programming assignment is due)
- The deadline to resubmit short quizzes is Wed 12/10 (2 weeks after the last quiz is due)
- The deadline to resubmit the long quiz is during Finals Week (exact date TBD).

These details will be provided on every assignment.

The Main Takeaway: Assignment deadlines are intentional, and you should submit your work by the deadline. You will be able to resubmit work that is marked “Missing” or “Partially Complete” after receiving feedback from me.

Collaboration Policy

Why Collaborate? Collaboration on all assignments and activities (except for the quizzes) is highly encouraged. It is often easier to work through problems with a thought partner, working with someone else brings a different perspective to the challenges, you might help each other out in complementary ways, and working together can be fun!

What's the Policy? When you collaborate, properly cite it in your work (see below). **You must write all of your own code for the programming assignments**, even if that means sitting next to a collaborator and typing the same thing. You have a lot of flexibility in naming variables, including print statements and comments in your code.

Code plagiarism is a real thing. Identical code is just as bad as copying and pasting entire paragraphs of an essay from another source. Just like when writing papers, there is a way to cite others' work in programs. If you collaborate with other students in the class or sought help from a tutor, add their names as comments, e.g.,

```
# This function was written with Jane and Rob for Lab 5
# I talked about the structure of this code with Marco
# Julie and I worked on the function post_network() together.
# I modified the convert() function from Lab 3
```

Remember, this type of collaboration is expected and encouraged! People who worked together should list each other as collaborators, since this will help us determine why some code might look similar.

Citing Previous Code from Class. If you copy your own code from previous assignments or labs, cite in the comments where you copied the code from (e.g., `# read_graph() function from Lab1.`). Anna will post solutions to the labs on Fridays; you are free to copy pieces of these solutions for future assignments, clearly state that it is from a posted solution (e.g., `# read_graph() function from posted Lab1 solution.`).

The Main Takeaway: Collaboration in class and on programming assignments is expected and encouraged! You should “cite” anyone you worked with on the programming assignments and any code you copied from a previous lab or assignment.

Online Resources & Generative AI Policy

There are many online resources for python, including python modules that have pre-packaged functions, the python standard library reference, and user help threads like StackOverflow. Generative AI is technology that is trained to generate text, images, or code from natural language prompts. Two examples of generative AI that you might have seen before are ChatGPT (which returns text based on prompts) and GitHub Copilot (which returns code based on prompts).

You may always use the resources linked from Moodle. This policy lays out the scope of what is allowed beyond the Moodle resources.

Programming Assignments

The fact is, online resources and generative AI might do a good job with some of the programming assignments. But the assignments are designed for *you* to learn about graph algorithms, which can only happen if *you* write the code. The course goals of this class include implementing graph algorithms and applying them to biological networks - if you use online resources or generative AI, you are not demonstrating what you have learned.

1. Do not use python packages that provide code for working with graphs (e.g., `networkx` or `igraph`) or other math/stats packages (e.g., `scipy` or `numpy`) unless otherwise directed. You *may* use `pandas` to manage datasets, but check with Anna before using any of the built-in functions.
2. You may look up basic Python syntax online (using either a search engine or, **new this year**, using generative AI). Some examples of basic syntax are “how do I structure a double FOR loop?” “How do I sort a list?” “How do I initialize a dictionary”?.
3. **New this year**, you may use generative AI to help debug your code. If you do, you must add a comment with the ChatGPT prompt you provided.
4. You may not look up the code for entire functions that are part of an assignment or lab (e.g., calculating the degree distribution of a graph or the shortest paths algorithm). While it may take some time for you to

write these functions yourself, they are important for the learning goals in the class.

When in doubt, “cite” your online resources in the comments of your code.

Quizzes

The quizzes are intended to be done on your own with your notes and the resources available on Moodle. They are an assessment of your understanding of biological networks and graph algorithms, so online resources (beyond the ones linked from Moodle) and generative AI are not allowed for quizzes.

Research Project

The research project, on the other hand, might have opportunities for using new packages and getting help with generative AI. We will have a class discussion about how to use generative AI in your research project to set the class policy then.

The Main Takeaway: Online Resources and Generative AI should not be used for generating entire functions, but they may be used to help understand basic python syntax and debug issues (with proper citation). Online resources should not be used for quizzes (beyond what Moodle provides), and we will discuss the role of online resources for the research project as a class.

Learning Environment

Bio331 brings together students from different majors to tackle inherently interdisciplinary topics, which is exciting but can also feel overwhelming at times. My goal is have a learning environment where students are motivated to learn the material without the fear of being incorrect or penalized for mistakes.

Feedback for the Instructor

Please give me feedback on anything in the syllabus or course, especially with respect to making Bio331 an inclusive learning environment. Let me know if anything makes you uncomfortable in class, if you would like more instruction on a topic, or if you are experiencing a hardship outside of class. Anonymous feedback is available on Moodle, though with a small class your comments may be identifiable.

Diversity & Inclusion

Bio331 is a combination of biology and computer science, and each field has been claimed to be free of racism and prejudice. This is simply not true. Historically, biological discoveries and advances in computer science have been dominated

by privileged voices, namely those of white men. Computational biology, while more recent of a field, also lacks diversity along many important axes (including race, gender, nationality, class, sexuality, religion, ability, etc.). To foster an inclusive learning environment:

1. I acknowledge the bias in course materials that stem from systemic privilege, and I aim to make Bio331 content more inclusive each time I teach the course.
2. I expect that students come to Bio331 with varying levels of knowledge in both biology and computer science, and I aim to provide course materials to help fill in gaps in both areas.
3. Many students will have personal circumstances that may affect their performance in the course. I will work with you to make adjustments to the course schedule as needed, and students should be encouraged to seek guidance on any anticipated or realized issues.
4. Mental and physical health is more important than attending all parts of class. If you miss multiple days of class, I will reach out to make sure you are okay.

Bio331 is a better course when students have a diversity of lived experience and previous knowledge.

Land Acknowledgement

As we come together to learn at Reed College, we acknowledge that the territory on which Reed stands is that of Indigenous and Native peoples. The Portland Metro area rests on traditional village sites of the Multnomah, Wasco, Cowlitz, Kathlamet, Clackamas, Bands of Chinook, Tualatin, Kalapuya, Molalla, and many other tribes who made their homes along the Columbia River (*Portland Indian Leaders Roundtable*). Native lands have been taken through hundreds of breaches of treaty laws, and in 1953 more than sixty tribes in Oregon were terminated by the federal government which removed millions of acres of land from Indigenous stewardship (*The Native American Community in Multnomah County: An Unsettling Profile*). This statement is one small step in acknowledging the history that brought us to reside on this land. Portland also has a vibrant and diverse Indigenous community and has the ninth largest urban Native American population in the US. We honor the present Indigenous communities in the area. Visit native-land.ca to explore an interactive map of Indigenous lands.

- **Start the assignments early.** The assignments are designed to be done over 1-2 weeks (and you may spend many hours debugging your programs). Read the assignments when they are assigned and follow the suggested guidelines to make sure you are on track to complete all the work.
- **This course is cumulative** – topics build upon each other throughout the semester. If you are having trouble with a concept or assignment, talk to me early before you find that you need that concept in a later assignment.

- **When in doubt, ask.** Asking for help is sometimes unintuitive, but a little clarification goes a long way. If you have the question, others will probably have the same question. Ask early so I can communicate the answers to the rest of the class. Also, sometimes I make mistakes! Your question might help me fix a typo.
- **When you don't even know where to start, ask.** This can be the toughest thing when working on problems that do not have a nice, clean solution - where to start? Coming to office hours when you don't even have a properly-formed question is still valuable.
- **Working collaboratively is fun!** Try to talk through complicated concepts with others, and make use of the group time in lab to get clarification. You can work with others outside of class on the programming assignments as outline in the collaboration policy.
- **Start the assignments early.** Really.

Change Log

The purpose of a change log is to document all major changes to some code/website/etc. Starting on September 2, all major changes will be documented in this change log. If you want to see *all* the changes made to this website since it was first created, take a look at the GitHub commit log.

9/1/2025

Adjusted the percentages in the Types of Work page to make programming assignments and short quizzes equal in contributions.

8/24/2025

Syllabus is finalized before the start of classes.